



International Credit Hours Engineering Programs
Computer Engineering and Software Systems
Agile Software Engineering

Lab 1: Git Basics & Local Repositories

Setup

1. Download and install Git from <https://git-scm.com>
2. Download and install VS Code from: <https://code.visualstudio.com/>

If you already have the above tools, no need to install them again.



Exercise 1: Preparing Your Project Environment

This exercise helps you prepare your development environment to use Git through the VS Code terminal.

1. Create a new empty folder on your machine named: agile-lab1
2. Open the Folder in VS Code
 - Open Visual Studio Code
 - Click: File → Open Folder → Select agile-lab1
3. Open the Integrated Terminal
 - From the top menu: Terminal → New Terminal (shortcut: Ctrl + `)
 - Make sure the terminal path is inside your project folder.
4. Check Git Installation by running the following command: **git --version**
If Git is installed, the version number should appear.
5. Configure Git (First Time Only)
Run the following commands (replace the highlighted text with your info)

```
git config --global user.name "Your Name"
```

```
git config --global user.email "your@email.com"
```
6. Initialize a Git Repository by running: **git init**
7. Verify Repository Initialization by running: **git status**
You should see a message indicating that an empty Git repository has been initialized.



Exercise 2: Working with a Local Repository

This exercise simulates a typical development scenario where you start a new feature.

Steps

1. Create two files:
 - One for code (ex: main.py)
 - One for notes (ex: notes.txt)
2. Check untracked status: **git status**
3. Write some code in the code file
4. Add the code file to the staging area: **git add main.py**
5. Check tracked status again
6. Commit the code file **git commit -m "Add initial code file"**
7. Explore commit history: **git log**
8. Update the code file and add it again
9. Write some information in the notes file
10. Add the notes file and commit it
11. Check status again
12. Check commit history (observe the commit hashes): **git log**
13. Check commit history using --oneline flag: **git log --oneline**